

Math Formulation: GAN Generator with Frozen Permeability Classifier and GRPO Training

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This is a working draft. Sections are meant to be extended as the architecture is refined.

1 Notation

- G_θ : generator policy, parameterized by θ . Produces a molecular graph m (atom/bond tensors, or a SELFIES sequence) from a latent sample $z \sim p(z)$.
- D_ϕ : discriminator, parameterized by ϕ . Outputs a scalar realism score $D_\phi(m) \in [0, 1]$, trained adversarially.
- C : the existing trained classifier ensemble (GIN/GAT/GINE), **frozen** — no gradient flows into C at any point. Outputs $C(m) \in [0, 1]$, the predicted probability that molecule m is BBB-permeable.
- $\text{Valid}(m) \in \{0, 1\}$: RDKit sanitization indicator.
- $\text{QED}(m)$, $\text{SA}(m)$: drug-likeness and synthetic accessibility scores, used as auxiliary reward terms.

2 Generator as a Stochastic Policy

Since molecule construction is discrete (choosing atoms, bonds, or SELFIES tokens) and passes through a non-differentiable sanitization step, G_θ is treated as a policy over a sequence of construction actions $a_1, \dots, a_T$:

$$\pi_\theta(m) = \prod_{t=1}^T \pi_\theta(a_t \mid a_{<t}, z) \tag{1}$$

The probability of the full generated molecule under the current policy is $\pi_\theta(m)$, and the ratio between the new and old policy (needed for the PPO-style clipped update in Section 5) is:

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid a_{<t}, z)}{\pi_{\theta_{\text{old}}}(a_t \mid a_{<t}, z)} \tag{2}$$

3 Discriminator Objective (Adversarial, Separate from the Classifier)

The discriminator is trained independently of C , using a standard (or WGAN-GP style) adversarial loss over real molecules $m^+ \sim p_{\text{data}}$ (from BBBP/B3DB) and generated molecules $m^- = G_\theta(z)$:

$$\mathcal{L}_D(\phi) = -\mathbb{E}_{m^+ \sim p_{\text{data}}} [\log D_\phi(m^+)] - \mathbb{E}_{z \sim p(z)} [\log (1 - D_\phi(G_\theta(z)))] \quad (3)$$

If using WGAN-GP instead (recommended for graph-structured outputs, following MolGAN):

$$\mathcal{L}_D(\phi) = \mathbb{E}_z [D_\phi(G_\theta(z))] - \mathbb{E}_{m^+} [D_\phi(m^+)] + \lambda_{gp} \mathbb{E}_{\hat{m}} [(\|\nabla_{\hat{m}} D_\phi(\hat{m})\|_2 - 1)^2] \quad (4)$$

where $\hat{m}$ is sampled along straight lines between real and generated molecules, as in the standard gradient penalty formulation.

D_ϕ is updated via gradient descent on $\mathcal{L}_D$; this is the *only* network trained with a standard differentiable loss in this whole architecture — everything downstream of the generator’s discrete sampling step needs policy gradients instead.

4 Reward Composition

For a generated, valid molecule m , define the composite scalar reward:

$$R(m) = \mathbb{I}[\text{Valid}(m)] \cdot (w_D D_\phi(m) + w_C C(m) + w_Q \text{QED}(m) + w_S \text{SA}(m)) + \mathbb{I}[\neg \text{Valid}(m)] \cdot R_{\text{invalid}} \quad (5)$$

where $w_D, w_C, w_Q, w_S \geq 0$ are weighting coefficients (hyperparameters to tune) and R_{invalid} is a fixed penalty (e.g. 0 or a small negative constant) for sanitization failures, so invalid molecules don’t silently drop out of the batch statistics used in Section 5.

Since C is frozen, $\nabla_\phi C(m) = 0$ always — the classifier only ever contributes a numeric reward value, never a gradient path back into its own weights.

5 GRPO: Group-Relative Advantage

For a given latent batch, sample a group of N molecules $\{m_1, \dots, m_N\}$ from the current policy $\pi_{\theta_{\text{old}}}$. Compute $R(m_i)$ for each using Section 4. The group-relative advantage for candidate i is:

$$A_i = \frac{R(m_i) - \mu_R}{\sigma_R + \epsilon}, \quad \mu_R = \frac{1}{N} \sum_{i=1}^N R(m_i), \quad \sigma_R = \sqrt{\frac{1}{N} \sum_{i=1}^N (R(m_i) - \mu_R)^2} \quad (6)$$

with a small ϵ for numerical stability. Note there is no learned critic/value network anywhere in this formulation — μ_R and σ_R are computed directly from the sampled group at each step.

6 PPO-Style Clipped Objective (with KL Penalty)

Using the advantage A_i from Section 5 and the policy ratio $r_t(\theta)$ from Section 2, the per-token clipped surrogate objective is:

$$\mathcal{L}_i^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) A_i, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_i \right) \right] \quad (7)$$

Averaged over the group, with a KL penalty against a reference policy π_{ref} (typically a recent checkpoint of θ , used to prevent the policy from drifting too far in a single update):

$$\mathcal{L}_G(\theta) = -\frac{1}{N} \sum_{i=1}^N \mathcal{L}_i^{\text{CLIP}}(\theta) + \beta \mathbb{E}_{m \sim \pi_\theta} \left[\text{KL}(\pi_\theta(\cdot | m) \parallel \pi_{\text{ref}}(\cdot | m)) \right] \quad (8)$$

θ is updated by minimizing $\mathcal{L}_G(\theta)$, typically over several epochs and minibatches drawn from the same sampled group before resampling a new one — the standard PPO reuse-of-batch pattern, carried over unchanged into GRPO.

7 Full Training Step (Algorithm Sketch)

1. Sample latent batch $z_1, \dots, z_N \sim p(z)$.
2. Generate molecules $m_i = G_{\theta_{\text{old}}}(z_i)$.
3. Filter/flag invalid molecules via RDKit; assign R_{invalid} where applicable.
4. Score valid molecules with D_ϕ (realism) and frozen C (permeability), plus QED/SA.
5. Compute composite rewards $R(m_i)$ (Section 4).
6. Compute group-relative advantages A_i (Section 5).
7. Run K epochs of minibatch clipped PPO updates on θ using A_i (Section 6).
8. Separately update ϕ (discriminator) via $\mathcal{L}_D$ (Section 3), typically on a different cadence (e.g. n_D discriminator steps per generator step, as is common in WGAN-GP training).
9. Set $\theta_{\text{old}} \leftarrow \theta$, repeat.

Open Items to Formalize Next

- Exact parameterization of $\pi_\theta(a_t | a_{<t}, z)$ — depends on whether the generator emits SELFIES tokens autoregressively or a one-shot atom/bond tensor (MolGAN-style), which changes what “action” a_t even means.
- Whether w_D, w_C, w_Q, w_S are fixed or annealed over training (e.g. starting weight on D_ϕ higher to establish validity/realism, then shifting weight toward C once the generator reliably produces valid molecules).
- Formal treatment of the discriminator/generator update ratio n_D and its effect on the reward’s non-stationarity from the generator’s point of view (since D_ϕ is a moving target, not frozen like C).